

Haseeb Autos — ERP & POS (Google Apps Script + Google Sheets)

Business: Haseeb Autos · Auto parts, car & bike decoration · Retail + Wholesale **Branches:** Sadiqabad City · Machi Goth · RYK Branch **Users:** 9 roles (Owner, Manager, Salesman, Purchase, Warehouse, Accountant, Cashier, Delivery, Other) **Stack:** Google Apps Script (backend) + Google Sheets (database) + Vanilla JS PWA (frontend) + Pluggable AI agent

Ye poora system **bina kisi server / hosting kharche ke** aapke Google account mein chalta hai.
Database = Google Sheets. Backend = Apps Script. Frontend = PWA (mobile + desktop + offline).

Is folder mein kya hai

```

haseeb-autos/
├─ apps-script/           # ← Yehi asli project hai (clasp push isi se hota hai)
│   └─ appsscript.json    # manifest (timezone, scopes, webapp)
│   └─ Code.gs            # doGet + API router (saari client calls yahan se
guzarti hain)
│   └─ Schema.gs          # 30+ sheets ki definition (single source of truth)
│   └─ DB.gs              # Sheets → database layer (cache, CRUD, number series)
│   └─ Setup.gs           # one-click setup: sheets + locations + 9 users +
settings
│   └─ Auth.gs            # login, session, RBAC (role + group permissions)
│   └─ Items.gs           # products, as-you-type search, nested pricing,
duplicates
│   └─ Inventory.gs       # per-branch stock, weighted avg cost, transfers,
audits
│   └─ Sales.gs           # POS billing, returns, holds, stock+ledger+commission
│   └─ Purchase.gs       # PO → GRN → purchase return
│   └─ Parties.gs        # customers, suppliers, ledgers, udhaar
│   └─ Payments.gs       # receipts, payments, expenses, cash sessions
│   └─ Accounting.gs      # □ v2.5 double-entry engine (vouchers, COA, P&L,
balance sheet, bank recon)
│   └─ Wallet.gs         # □ v2.5 EasyPaisa / JazzCash gateway (REST v4 / pp_
API)
│   └─ Orders.gs         # □ v2.5 order book – mobile order taking → invoice
│   └─ Reports.gs        # dashboard KPIs + 10 reports
│   └─ Config.gs         # ⚙ v2 config engine – Settings UI isi schema se banti
hai
│   └─ Warehouse.gs      # 📦 v2 bins/racks, put-away, bin transfer, cycle
count, lots
│   └─ Notifications.gs  # □ v2 alert engine (low stock, udhaar ageing, PO,
expiry)
│   └─ Seed_Products.gs  # 📦 71 asli Flamingo products (list.pdf + barcode.pdf)
│   └─ AI.gs             # AI agent (Gemini / OpenAI / OpenRouter / Mock) + 17
tools
│   └─ Barcode.gs        # barcode generation, QR payloads, label templates
│   └─ OfflineSync.gs    # offline catalog pull + queued actions sync
│   └─ Audit.gs          # before/after audit trail
│   └─ Utils.gs          # helpers (money, dates, fuzzy search, hashing)
│   └─ *.html            # PWA frontend (Index + Styles + 14 JS modules)
│       └─ App_UI2.html   # nested tabs, offcanvas, modal, SmartTable, form
builder
│       └─ App_POS2.html  # POS v2 – product CARDS, category rail, cart, split
payment
│       └─ App_Masters.html # Items / Customers / Suppliers – offcanvas sub-tabs
│       └─ App_Inventory2.html # Inventory + Warehouse (count sheet, bins, lots)
│       └─ App_Config.html  # Settings & Configuration (sab kuch editable)
│       └─ App_Dashboards.html # chart engine + AI dashboards + shop open/close
│       └─ App_QR.html      # offline QR encoder (koi CDN nahi)
├─ demo/
│   └─ index.html         # self-contained demo (mock data) – preview ke liye
│   └─ mock.js            # fake backend (sirf demo ke liye)

```

```

├─ tools/
│   ├── build_demo.py      # apps-script/*.html → demo/index.html
│   ├── check.sh           # syntax check (.gs + templates)
│   ├── deploy.sh          # clasp push one-command
│   ├── package.sh         # zip bundle banata hai
│   ├── parse_products.py  # list.pdf + barcode.pdf → products.json (catalog
import)
│   ├── gen_seed.py        # products.json → Seed_Products.gs + demo/products.js
│   └── test_logic.js       # 64 business-logic tests (real .gs backend, mock
Sheets)
│   ├── test_qr.js         # QR encoder matrix dump
│   ├── verify_qr.py       # QR ko reference encoder se compare
│   ├── smoke.js           # headless UI smoke test (har screen + har tab)
│   ├── build_pdf.py       # README + DEPLOYMENT → release/Haseeb-Autos-Guide.pdf
│   └── package.sh         # release/haseeb-autos-v2.1.0.zip bundle
└─ docs/ (DEPLOYMENT.md)  # step-by-step deployment

```

NEW v2 mein kya naya hai

- 1. Kuch bhi hardcoded nahi — sab Settings ► Configuration se** Har label, module, field, tab, price tier, category, filter, workflow, permission, print template aur number-format **Sheets mein stored** hai aur UI se edit hota hai: - **Business / POS / Inventory / Trade / Modules / Automation / Security groups** — har group ke andar **sub-tabs** (Settings screen khud `Config.defs()` schema se banti hai) - **Custom fields** — Item / Customer / Supplier / Sale / Purchase par apne fields (text, number, date, select, switch) aur form-tab choose kar ke - **Menu builder** — sidebar ka har item (label, icon, order, permission, parent) editable - **Print templates** — receipt/label JSON template (barcode/QR on/off, size, footer) - **Lists** — Categories, Brands, Units, Tax codes, Warehouses sab editable - **Modules on/off**, theme (colour, density, font size), automation, security
- 2. POS = product CARDS** — category rail, as-you-type filter (name/code/barcode/brand/model), stock badge, favourites, recently-sold, held bills, qty pad, **split payment**, **camera barcode/QR scan** (mobile), offcanvas cart (mobile), F2/F4/F8/F9 shortcuts, Ctrl+K command palette.
- 3. Warehouse first-class module** — warehouses, bins/racks, put-away, bin-to-bin transfer, bin-wise stock, count sheet (system vs counted → adjustment), lots/expiry.
- 4. AI-driven interactive dashboards** — pure-SVG charts (donut, bars, area, sparkline, gauge, progress, heat): Main dashboard + Customers · Receivables · Payables · **P&L** · Inventory · Alerts, aur **Shop Open/Close** (denomination counter + variance).
- 5. Har jagah quick filter** — SmartTable har table mein: as-you-type filter **har field** par, multi-facet filters (category/brand/status/branch/date/stock), column picker, density, **saved views (filter presets)**, CSV export, sticky header, internal scroll (**bottom scroll nahi**).
- 6. Deep links** — kisi bhi jagah se drill-down: `items?open=ID` , `parties?tab=customers&open=ID` , `sales?open=ID` , `purchase?reorder=true` — har link kaam karta hai (88 UI checks se verified).

7. Configuration backup/restore — Settings ► Data ► Backup/restore se poora setup (settings + lists + menu + custom fields + templates) JSON mein export/import.

8. Offline QR + barcode — built-in QR encoder (internet/CDN zaroori nahi), Code39 + EAN-13 barcodes, labels aur receipts dono par.

9. Verified calculations — 64 automated logic tests: weighted-average costing, COGS, tax (exclusive/inclusive, discount-before-tax), ledger, transfers, adjustments, guards.

v2.1 mein kya naya hai

1.  Auto Reorder engine (`Reorder.gs` + `App_Reorder.html`) — Settings ► Automation ► *Purchase reorder rules* se sab kuch configurable (method `MIN_MAX` / `REORDER_POINT` / `VELOCITY` , lookback days, cover days, lead time, safety stock, pack rounding, min order value, auto-PO `DRAFT` / `APPROVE`) :
- **Suggestions** tab: velocity, on-hand, on-order, suggested qty, value, priority (`OUT_OF_STOCK` / `CRITICAL` / `BELOW_REORDER` / `LOW_COVER` / `REPLENISH`) - **Rules & preview** tab: live calculation preview (formula dikhata hai) - **Generated POs** tab + *Create POs* → supplier-wise grouped purchase orders - Deep link: `purchase?reorder=true` → PO draft with lines

2.  Bilingual engine (`EN` / `Roman Urdu` / `اردو`) (`Lang.gs` + `App_Lang.html`) - Nav ki `#EN` button: click = cycle `EN` → `RO` → `UR` (Urdu par **RTL** layout), double-click / right-click = **translation editor** - `Translations` sheet mein har key ke 3 columns (en / roman / ur) — UI se edit, **AI se khali keys auto-fill** (`Lang.auto` , Gemini/OpenAI/OpenRouter, 25 keys per batch) - Koi key missing ho to English fallback — kabhi blank screen nahi

3.  POS camera scanner —  `Scan` button (native `BarcodeDetector` : Code128/39/EAN-13/EAN-8/UPC/QR) + manual entry fallback + beep on hit.

4. ☆ Saved views — SmartTable ke `☆ Views` se filter/sort/columns preset save, apply, delete (per screen, `SavedViews` sheet).

5.  Backup / restore — Settings ► Data ► Backup: poora config (settings + custom fields + menu + templates + 5 lists) JSON export/import — machine change ya naye branch setup mein 10 second ka kaam.

6. ✂ Dashboard quick actions — New sale · Add item · Receive udhaar · Shop close · Reorder stock · P&L · Warehouse · Ask AI (sab deep links).

7.  Real PWA assets (static hosting) — `manifest.webmanifest` , `sw.js` , `icon-192/512.png` (cache-first app shell, network-only API). Apps Script `/exec` scope mein inline data-URI manifest; static hosting par asli installable PWA.

8. Offline QR everywhere — `QR.dataUrl()` ab pehle built-in offline encoder use karta hai (network sirf last fallback).

□ Apps Script se deploy — mukammal guide

APPS_SCRIPT_DEPLOY_GUIDE.md — zero se live tak, Roman Urdu mein: clasp se push, copy-paste method, `setupAll()`, Web app deployment, users/roles, AI key, **daily triggers**, mobile PWA install, updates aur troubleshooting table.

✂ 10 minute mein live kaisy karein (short version)

1. **Sheet + code create karein** — script.google.com → New project → **Setup.gs** etc. paste karein (ya clasp use karein).
2. **Setup.gs** mein function select kar ke **setupAll()** Run ► → Google authorize karein. → Ek naya Spreadsheet "Haseeb Autos - ERP Database" ban jata hai jisme **30+ sheets**, 3 branches aur **9 users** ready hain.
3. **Deploy ► New deployment ► Web app** → Execute as: **Me** · Access: **Anyone** → URL copy karein.
4. URL kholein → **owner / admin123** se login karein → **Password turant badlein**.
5. AI chalana hai to: Project Settings ► Script Properties ► **AI_API_KEY** = apni Gemini key.

Poora detail: [DEPLOYMENT.md](#)

□ Default accounts (pehli login ke baad password sab se pehlay change karein)

Username	Role	Password	Kya kar sakta hai
owner	OWNER	admin123	Sab kuch (full admin)
manager	MANAGER	manager123	Sales, items, purchase, reports, users
sales1	SALESMAN	sales123	POS billing, items dekhna, customers
purchase	PURCHASE	purchase123	PO, GRN, suppliers, stock
warehouse	WAREHOUSE	ware123	Stock, transfers, adjustments, GRN
accountant	ACCOUNTANT	acc123	Reports, payments, ledgers
cashier	CASHIER	cash123	POS + cash drawer
delivery	DELIVERY	del123	Sirf dekhna (sales/items)
other	OTHER	other123	Minimum read-only

□ Features (jo aapne demand kiye thay)

1. POS (billing)

- **As-you-type search** — jaisay hi type karein (**brk** , **corolla** , **HLG-001**) results turant, bina Enter ke
- **Barcode / QR scanner** support (scanner keyboard ki tarah type karta hai, Enter pe direct add)
- Cart: qty +/–, price edit, line discount, min-price guard
- Split payment (Cash + Card + JazzCash + Easypaisa + Bank + Cheque)
- Udhaar sale with credit-limit check · Change calculate · Hold/Resume bills
- Thermal receipt 80mm print with barcode + Urdu header
- Keyboard: **F2** search · **F4** customer · **F8** hold · **F9** pay · **Esc** clear

2. Inventory (3 branches)

- Har branch ka alag stock · weighted-average cost · rack/shelf
- Stock ledger (har movement ki record) · quick count
- Branch-to-branch transfer (send → in-transit → receive)
- Stock adjustment: **Draft → Post → Reverse** (audit reversal)
- Low stock alerts + reorder suggestions with estimated cost
- Negative stock block (setting se allow kiya ja sakta hai)

3. Items (products)

- Auto-parts fields: brand, part type, **make / model / year range / engine / chassis**
- Multiple barcodes (alternate barcodes) · auto barcode generate
- 3 price levels: cost · retail · wholesale + **customer-type based nested pricing** (qty slabs)
- Min-price guard (salesman is se neechay nahi ja sakta)
- CSV import/export · duplicate finder · bulk status change · **barcode label printing** (Code39, offline)

4. Purchase

PO (draft → approve) → **GRN** (stock in + cost update + supplier ledger) → Purchase return Budget limit check · pending qty tracking · on-credit purchase

5. Customers / Suppliers

- Udhaar ledger with running balance · credit limits · customer types (discount tiers)
- Membership points (Rs 1000 = 1 point) · receive payment / pay supplier
- Receivables & payables reports

6. Reports

Dashboard (8 KPIs + 14-day trend + top items + low stock + payment mix) · Sales · Item-wise · Category-wise · Profit & loss · Stock valuation · Low stock · Receivables · Payables · Cash book — sab **CSV export** ke sath

7. Users & Security (RBAC)

- 9 roles × permission keys (e.g. `pos.discount` , `items.price.edit` , `reports.financial`)
- Groups with extra permissions + sale-return limit + data-view limit (days)
- **Audit log**: har write pe before/after snapshot, diff viewer
- Password SHA-256 + salt + pepper · session expiry 12h · branch-level data access

8. 🤖 AI Agent (business automation)

Frontend chat (Ctrl+K) + backend agent with **17 tools** jo aapke real data par chaltay hain:

Tool	Example sawal
<code>search_items</code> / <code>price_check</code>	"HLG-001 kitne ka hai?"
<code>get_item_stock</code>	"Corolla brake pad kitna stock hai?"
<code>today_sales</code> / <code>sales_summary</code>	"Aaj ki sale kitni hui?"
<code>top_selling_items</code>	"Is month top 5 parts?"
<code>low_stock_list</code> / <code>reorder_suggestions</code>	"Kya mangwana chahiye?"
<code>customer_balance</code> / <code>receivables</code>	"Ahmad ka udhaar kitna hai?"
<code>profit_report</code> / <code>cash_position</code>	"Is mahine ka profit?"
<code>draft_sale</code> / <code>draft_purchase_order</code>	"2 brake pad + 1 oil ka bill bana do"

Safety: AI sirf whitelisted tools chala sakta hai, user ki RBAC permissions ke ander, aur write tools **draft** banatay hain (post nahi) jab tak `aiCanWrite = true` na ho. Provider: **Gemini** (default) / OpenAI / OpenRouter — key sirf Script Property mein.

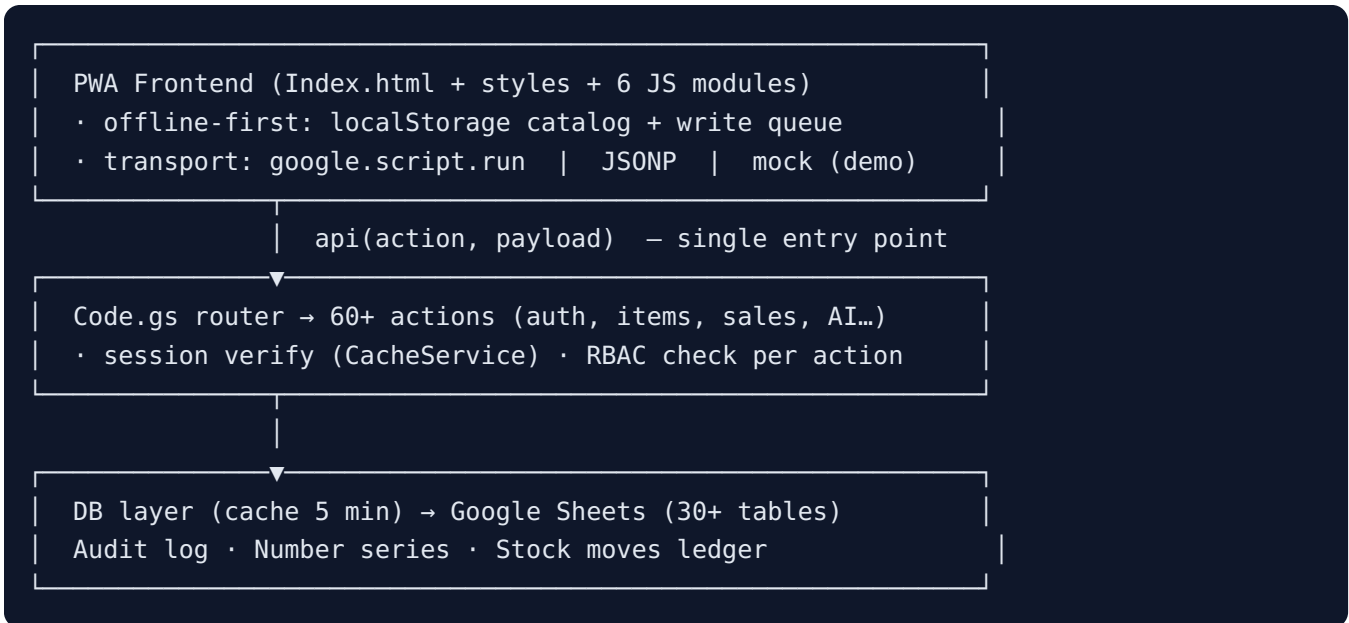
9. PWA + Offline

- Install karein mobile/desktop par (home screen icon, standalone window)
- **Offline billing**: internet jane par bhi POS chalta hai (local catalog + write queue), wapas online aate hi **auto-sync** with duplicate protection
- Mobile-first UI: bottom tab bar, touch-friendly cards, 80mm print, camera/scan support

10. Extras

Multi-language toggle (**English ⇌ Roman Urdu**) · Dark/Light theme · branch switcher · global search (items/invoices/customers) · notifications · audit trail · Drive backup · data purge/reindex utilities

🏗️ Architecture (ek nazar mein)



Transaction safety: Sale create = stock out + sale items + customer ledger + payment + commission + audit — ek hi server call mein, ya to sab ho ga ya kuch nahi.

📄 Developer commands

```
bash tools/verify.sh          # ☆ SAB 13 GATES EK SATH (ye chalaayein)
bash tools/check.sh           # syntax check (.gs + html templates)
python3 tools/build_demo.py   # demo/index.html rebuild
node tools/test_logic.js      # business-logic tests (real backend, mock Sheets)
node tools/smoke.js           # headless UI smoke test – har screen + har tab
node tools/audit_schema.js    # Sheets/DB schema health (silent data-loss check)
node tools/audit_layout.js 1440 900 --dark --overlays # asli browser: layout/contrast
node tools/verify_punchlist.js # 16 required cheezein – acceptance gate
node tools/test_qr.js && python3 tools/verify_qr.py # QR encoder verification
bash tools/deploy.sh          # clasp push (agar clasp installed hai)
bash tools/package.sh 2.5.2   # release/haseeb-autos-v2.5.2.zip + PDF guide
```

Apni product list CSV se import (asli product master)

```
python3 tools/import_products_csv.py path/to/export.csv # CSV → tools/products.json (+DQ report)
python3 tools/import_products_csv.py --report-only # sirf analyze, kuch likhay baghair
python3 tools/import_products_csv.py --price-mode swap # jahan wholesale>retail, swap kar dein
python3 tools/gen_seed.py # products.json → Seed_Products.gs + demo/products.js
python3 tools/build_demo.py # demo rebuild
python3 tools/gen_product_report.py # release/product-import-report.html
```

Phir Apps Script editor mein `seedRealProducts()` chalayein — **ya** deployed app mein Settings ► Data & tools ► Maintenance →  **Seed real products.**

Expected CSV columns (aapke export jaisay): `Product Code, Line Item, Category, Sub Category, Conversion Factor, Supplier, Cost Type, Purchase Type, Made in, Product Group, Model, Brand, Product Name, Size, Color, Cost Price, Wholesale Price, Retail Price, Profit(%)`, `Difference, Barcode`.

Saari suites **green** hain — `bash tools/verify.sh` → **ALL GATES GREEN** (check 29 blocks · schema 0 problems · logic **581/0** · ui 0 · smoke **308/0** · tabs **483/0** · align **36/0** · punch **16/16** · **117/117** · 5 × layout **0 problems** — desktop/mobile/dark/overlays).

⚠ Important notes

- **Google Sheets limits:** ~10M cells. Is schema se lagbhag 1M+ transactions aa jayen gi; us ke baad purana data archive karein (Settings ► Data ► Purge) ya Firebase/SQL par migrate karein.
- **Barcode format:** Code39 (built-in, offline, har scanner par chalta hai) — catalogue ke 71 Flamingo products mein asli EAN-13 bhi maujood hain; EAN-13 generate bhi ho sakta hai.
- **QR** ab **built-in offline** encoder se banta hai — internet/CDN ki zaroorat nahi (`App_QR.html` , reference encoder se byte-exact verified).
- **Service worker** Apps Script `/exec` scope mein register nahi ho sakta, is liye offline continuity app ke apne catalog + queue se handle ki gayi hai (installable PWA ke liye static hosting mode dekhein — `DEPLOYMENT.md` ► Mode B).

Version: 2.5.0 · **Built for:** Haseeb Autos · **License:** Aapki property — freely modify & resell.

v2.3 features guide (WhatsApp/SMS · price-list import · loyalty · export · Firebase)

Roman-Urdu step-by-step guide for everything added in v2.3: [FEATURES V23 GUIDE.md](#)

- WhatsApp (free wa.me / Meta Cloud API / Twilio), SMS, Email + outbox & retry
- Supplier price-list import (CSV paste, Gmail attachment, auto column detection, margin guard)
- Loyalty points: earn formula, POS redemption, cap/min, expiry, manual adjustment
- Export & share: Print/PDF, CSV, Excel (xls + Drive xlsx), native share
- Firebase migration: tier assessment, JSON export, Firestore import, 7-step cutover

v2.2 feature guide (receipts · payment methods · AI agent · dashboards · verification)

Roman-Urdu guide for everything added in v2.2: [FEATURES V22 GUIDE.md](#)

- Professional receipts/invoices: thermal 58/80/110mm + A3/A4/A5/Half/Letter, ESC/POS, visual designer
- Invoice previous balance (ledger-accurate)
- Real payment methods: cash, card, bank, JazzCash, EasyPaisa, Raast, cheque lifecycle, credit — fee/FED/settlement formulas
- Popup alignment: offcanvas right-edge, modals centered (audited)
- AI agent configuration under Automation (provider/model/key/guardrails/tools/presets/test)
- Professional dashboards: donut, gauge circles, multi-series lines, bars, progress, sparkline
- Verification suite: `bash tools/verify.sh` (check · logic · ui · smoke · tabs · align)

Haseeb Autos ERP/POS — v2.5.2 (real product master)

Sab kuch **settings-driven** — koi feature, koi model naam, koi paper size hardcode nahi. Har naya toggle Settings ► (mutaliqa tab) mein edit ho sakta hai.

1. AI agents — ab koi model name code mein nahi

"you still hardcoded ai agents model name, gemini 1, 2 are depreciated..."

Pehle	Ab
<code>AI_MODEL_CATALOG</code> mein model ids code mein likhe hue the	Catalog Settings (<code>ai.modelCatalog.<PROVIDER></code> JSON) ya live discovery (<code>/models</code> endpoint) se aata hai
<code>AI_DEFAULT_MODEL</code> map (hardcode)	<code>AI.defaultModel(provider)</code> — Settings override → catalog se behtareen (stable + free-tier + sab se naya version)
<code>AI_DEFAULTS.aiModel = 'gemini-2.0-flash'</code> (retired)	<code>aiModel: '' = auto</code> (provider se latest/free model khud chun leta hai)
<code>cfg.model \ \ 'gpt-4o-mini'</code> (deprecated fallback)	Khali = auto; na mile to UI "Refresh models" ya custom id ka kehta hai
OpenAI family filter code mein	<code>aiOpenaiPrefixes</code> setting (comma list)

- Naye settings: `aiPreferFree` (free tier ko tarjeeh), `aiModelCacheMins`, `ai.modelCatalog.GEMINI/OPENAI/OPENROUTER`, `aiOpenaiPrefixes`.
- Frontend se bhi `AI_DEFAULT_MODEL_HINT` map hata diya — provider badalte hi recommended model backend se aata hai.
- **Regression guard:** `tools/test_logic.js` ab source scan karta hai — kisi bhi `.gs` / `.html` ki **code line** (comment ke ilawa) par vendor model id ho to test fail. Demo data (`DEMO_MODEL_CATALOG`) alag block mein hai jo UI se replace ho sakta hai.

2. Charts — bars/donut/gauge (industry best practice)

- Bar charts ab **HTML rows** (`.bars-html`): label (ellipsis) | track/fill | compact right-aligned value — **text/value overlap structurally impossible**. Auto-delegate jab rows > 8 ya label > 12 chars ya horizontal mode.
- `preserveAspectRatio="none"` kahin nahi (yehi overlapping text ki wajah thi).
- Donut 120–170 px, gauge ≤ 118 px, bar chart height ≤ 220 px — har jagah clamp.
- 5 screens (dashboard / insights / reorder / accounting / warehouse) × 4 assertions ka smoke gate.

3. Professional invoice (print preview) □

- **A4 / A5 / Half-A4 tax-invoice layout:** letterhead (logo, business name EN + Urdu, tagline, address, phone/email, NTN/STRN), **TAX INVOICE** badge, **Bill to** box, meta grid (invoice no, date, salesman, PO/reference, terms, due date, branch).
- Items table: #, description + part no, HSN, Qty, Unit, Rate, Disc, Tax, Amount — zebra rows, header repeat on page break, **break-inside: avoid**.
- **Amount in words** (PK numbering: lakh / crore) — **Rupees Five Thousand Four Hundred Forty and Fifty Paisa Only**.
- **Bank / payment details** block (Settings ► Business ► Bank & invoice).
- **Account summary:** Previous balance → This invoice → Paid now → **Closing balance** (2500 + 1000 = 3500 wala hisaab).
- Signature lines, terms & conditions, barcode/QR, page-break rules.
- Print dialog mein **quick formats**: □ Thermal receipt · □ A4 tax invoice · □ A5 invoice · □ Half A4.
- Naye settings: `bankName/bankTitle/bankAccount/bankIban`, `invoice.showBank`, `invoice.showWords`, `invoice.showSignature`, `invoiceTerms`, `print.decimals`, `strn`, `currencyWord`.
- Sample output: `release/invoice-sample-a4.html`, `release/invoice-sample-thermal.html`, `release/day-report-sample.html`.

4. Previous balance har jagah nazar aaye □

- **POS customer bar:** **Prev 2,500 → 3,500 (bill 1,000)** live.
- **Payment dialog:** Account summary block (previous / this invoice / paid / closing).
- **Customer & Supplier detail:** ledger-se computed breakdown — Opening + Billed — Paid = Current (naya route `parties.balance`).
- **Ledger tab** KPIs: Opening · Billed · Received · Closing.

5. Purchase orders & GRN □

- **PO form:** professional column layout (Item | Stock | Qty | Rate | Amount), per-line stock, received/pending, detail strip (status, lines, ordered/received/pending, budget, GRN received count), **Approve** action.
- **GRN form:** "Load from PO" dropdown — sirf wo POs jin ki pending lines hain (`PO-SDQ-01001 · Karachi Auto Traders · 6 items · pending 80`); select karte hi pending lines, supplier aur branch load; is ke baad manual item bhi add kar sakte hain. Columns: Item | Ordered | Received | Now | Cost | Amount.

6. Auto-reorder □

- "Generated POs" tab ab har PO ke sath **"6 items"** dikhata hai (backend `itemCount` + `lines`).
- "Reorder ke liye products" card ab **Order qty, Est. cost** aur **Supplier** dikhata hai (pehle — tha kyunki `suggestedQty` /supplier fields missing the).

- `Items.reorderSuggestion` ab supplier name aur suggested qty (settings `reorderTargetMultiplier`) ke sath deta hai.

7. Shop Open/Close + Day/Shift Closing Report

- Header ab **Open / Closed pill** (colored dot, tooltip mein session no + expected cash), auto-refresh har 2 minute.
- Pill par click → **quick dialog**:
- Band ho → opening cash (quick chips 500/1000/2000/5000) + note → **Open shop**.
- Khuli ho → day summary (session/duration, opening, cash in, cash out, card/wallet, **expected in drawer**, txns)
 - **Close shop / Cash in / Cash out / Full screen**.
- **Naya: Day / Shift Closing Report** (`shop.dayReport`) — पूरी shift ki tasveer:
- Header: business (EN + Urdu), branch, phone, NTN · session no · cashier · opened/closed · duration
- KPIs: bills · gross sales · returns · udhaar (credit) · card/wallet · cash sales
- **Cash drawer reconciliation**: opening + cash sales — cash out — expenses = **expected** → counted → **variance**
- Payments by method · Expenses by head · **Top items** · **Hour-wise business** · Cash movements (time/type/reason/amount)
- Notes + cashier & manager signature lines
- Report actions: **Print** · **PDF** (server `exports.pdf` → Drive PDF + shareable link; offline ho to browser print → "Save as PDF") · **WhatsApp** (free wa.me link — PDF URL + summary) · **CSV** (Excel-ready).
- Report khulega: shop hero se, purani sessions ki table row se, aur **shop close karte hi khud-ba-khud**.
- Naya generic route `exports.pdf` — koi bhi HTML (invoice/report/document) Drive PDF ban kar share ho sakta hai.

8. Professional settings & toggles

- Har switch ab **settings-row** mein: label + hint baen taraf, modern 44×24 switch daen taraf, **On/Off** state text, focus ring, disabled state, press animation (koi chhota "ON/OFF" text ab track ke andar nahi).
- Nav: Users & Security ► Permissions matrix (43 permissions × 9 roles) + naya **per-user rules editor**.

9. Users & Security — per-user rules

- User form mein **"Rules — kaun se actions allow / disallow hain"**: har permission ke liye **tri-state** `Inherit (role)` → **Allow** → **Deny**, search + "Sab inherit".
- Backend: `Auth._buildSession` per-user `extraPermissions` (allow) aur `deniedPermissions` (deny) lagoo karta hai; `users.create` / `users.update` in fields ko save karte hain.

10. Product cards (POS) □

- Image **strict 1:1** (`aspect-ratio: 1/1` , width 100%), `pos.cardImageFit` = contain/cover setting.
- Card par **supplier ka naam** (🏢) — `pos.showSupplierOnCard` .
- **Hover quick-view popover** (350 ms): code, supplier, stock, retail, wholesale, cost, rack — `pos.hoverQuickView` setting; ⓘ button se full record (modal) pahle ki tarah.

11. Transactions & Expenses / Accounts / Banks □

- **Expense voucher** ab professional: paid-from **cash/bank account**, **bill/reference no**, **tax amount**, **attachment URL**, status (POSTED / PENDING / REJECTED).
- **Approval workflow** (settings-driven): `expenseApproval` + `expenseApprovalOver` — limit se bada expense **PENDING** rehta hai aur approve hote hi voucher post hota hai. Table mein Status column, row actions (Approve / Reject / Attachment) aur "Sab approve karein" strip.
- **Expense detail drawer**: amount, method/account, status, paid to, bill ref, tax, note, attachment.
- Naye settings: `expenseApproval` , `expenseApprovalOver` , `expenseCategories` .
- Banks tab: bank accounts KPI + **Bank book** (opening / in / out / closing + running balance) + **Reconciliation**.

12. Cash book — opening ab hardcoded nahi □

Pehle `Reports.cashbook` mein `opening: 0` hardcoded tha — yani har report sirf usi range ka net dikhata tha, pichhla balance kahin nahi. Ab poori tarah settings-driven hai:

- **Opening** = `cashbook.openingCash` (settings ► Trade & Accounts ► Accounts) + us tareekh (`cashbook.openingAsOf`) se period start tak ki **saari** purani cash movements. As-on date khali ho to poori history ka net le leta hai — koi hardcoded zero nahi.
- **Running balance**: har entry ke saath `balance` (opening → in → out → balance) — bank statement ki tarah.
- **Day-wise cash book**: date · opening · in · out · net · closing (har din apni opening/closing ke saath).
- **Method filter** (`cashbook.method`): `ALL` ya koi ek method — CASH chunien to asli *cash book*.
- **Expenses** cash-out mein shamil (`cashbook.includeExpenses` setting se band ho sakti hain).
- Report screen par: 4 KPIs (Opening / Cash in / Cash out / Closing), **"Opening kahan se aaya"** audit strip (settings balance + purani entries + ⚙ Settings shortcut), method-wise table, day-wise table, entries table — sab **CSV / PDF / WhatsApp** export ke saath.

13. Code hygiene & docs

- Murda (dead) duplicate **Settings** screen `App_Screens2.html` se hata di — ab sirf ek hi professional **Settings & Configuration** screen hai (`App_Config.html`).
- `release/Haseeb-Autos-Guide.pdf` dobara **v2.5.1** ke mutabiq banaya gaya (README + **v2.5.1 changelog** + deploy guide + deployment + features guide).

14. Punch-list acceptance gate — har requirement ab machine-verified

`tools/verify_punchlist.js` (naya 7-wa gate, `bash tools/verify.sh` mein shamil) aap ki 16 requirements ko 1:1 automated checks mein badal deta hai — **static** (source code) + **live** (jsdom ke andar demo app chalakar):

#	Requirement	Kya check hota hai
1	AI agents — koi hardcoded model name	14 vendor model names ka scan (code mein kahin nahi) + <code>AI.catalog / discoverModels / defaultModel</code>
2	Charts — bars overlap nahi, donut/gauge size band	bars HTML rows (<code>.bars-html .bh-row</code> : label + value alag column), <code>SIZING</code> clamp (donut $\leq 170\text{px}$)
3	Professional print preview	A4/A5/Half-A4/thermal layouts, <code>Previous balance</code> , amount-in-words, ESC/POS
4	Customer previous balance har jagah	<code>Parties.summary</code> (opening + billed – paid), POS "Previous/Closing balance", reports "Pichhla baqaya"
5	Suppliers — wahi payable hisaab	supplier convention (opening + purchase – paid), supplier-wise payable report
6	Transactions/Expenses/Accounts/Banks	expenses Status + approval, chart of accounts, bank book (opening/in/out/closing), wallets tab
7	EasyPaisa / JazzCash real integration	HMAC helpers, <code>ppmpf_1</code> payer mobile, sandbox/live hosts, inquire (<code>epStatus / jcStatus</code>), payer+receiver mobile fields
8	PO detail + GRN mein PO load	<code>.po-line</code> line editor, <code>loadPoIntoGrn</code> , PO selector GRN form mein
9	Auto reorder — Orders to verify	<code>suggestedQty</code> / <code>supplierName</code> / multiplier + rows render (NaN-free)
10	Shop Open/Close + closing report	header <code>.shop-pill</code> , <code>shop.dayReport</code> , <code>exports.pdf</code> , Print/PDF/WhatsApp/CSV
11	Export PDF / Excel / CSV + WhatsApp	<code>Exports.csv/sheet(xlsx)/report/htmlToPdf</code> + free <code>wa.me</code> share
12	Settings & toggles professional	<code>.f-row-switch</code> rows + modern <code>.switch</code> + On/Off state
13	Warehouse app	

#	Requirement	Kya check hota hai
		<code>countCreate/countPost/binTransfer/audit/putaway</code> + Count sheets/Audit/Bins tabs
14	Mobile order-taking app	<code>registerScreen('takeorder')</code> + order → convert → advance
15	Product cards 1:1 + quick view	<code>aspect-ratio:1/1</code> on <code>.pcard</code> <code>.pc-img</code> , <code>.pc-sup</code> supplier line, <code>.pc-pop</code> hover popover
16	□ Users & Security rules	<code>extraPermissions</code> / <code>deniedPermissions</code> + tri-state rules editor (Inherit/Allow/Deny)

Har run par `release/punchlist-report.html` ban jata hai (khulne layaq report: requirement, kyun zaroori hai, aur har check ka ✓/✗). Ab koi bhi feature tootay to ye gate fail hoga.

15. Design system & accessibility — asli browser mein 358 → 0 maslay □

jsdom layout nahi karta, is liye UI ke asli masle nazar nahi aate. Is liye ab ek **real-browser audit gate** hai (`tools/audit_layout.js` , Chromium/puppeteer) — har screen ko 1440×900 aur 390×844 par chalakar asli geometry napta hai.

Pehle run par **358 problems** miley — sab fix:

Masla	Pehle	Ab
Contrast (WCAG AA)	brand <code>#ff6a00</code> text/button sirf 2.87:1 (AA = 4.5:1), ok/warn/err/info sab 3:1 ke qareeb	Naye tokens: <code>--brand-text:#b7410e</code> (text), <code>--brand-strong:#c2410c</code> (solid button, safed text ke sath 5.1:1); <code>--ok/#15803d</code> , <code>--warn/#a16207</code> , <code>--err/#b91c1c</code> , <code>--info/#0369a1</code> , <code>--muted/#556274</code> , <code>--muted-2/#5f6d80</code>
Chart palette	safed label orange/green fill par 2.76:1	palette darken (<code>App_Charts.PAL</code> , <code>App_Dashboards.PALETTE</code>) — har tone safed label ke sath ≥4.5:1
Tap targets	chips 21px, seg/tabs 29-30px, table inputs 25px	sab ≥ 32px (mobile best practice ke qareeb), FAB 56px, card “+” 30px
Chhupe hue tabs	Settings ke 12 tabs hidden horizontal scroll mein	<code>.tabbar-l1</code> wrap — sab tabs nazar aate hain
Mobile overflow	POS search bar 390px par 27px bahar	<code>.pos2-search</code> wrap + shrinkable search box, <code>.card-head</code> <code>.actions</code> wrap
Kati hui text	Items table mein “Flamingo · DEGREASER FOAMING...” bina ellipsis	ellipsis + <code>title</code> tooltip

Audit har run par check karta hai: **overflow** (viewport se bahar), **clipped text**, **text overlap** (jaise purani bar charts), **tap target <32px**, **WCAG AA contrast** (translucent backgrounds ko sahi se composite karke), **duplicate ids**, **horizontal page scroll**.

1440×900 → PROBLEMS: 0 1024×768 → PROBLEMS: 0 390×844 → PROBLEMS: 0

16. Dark theme + modals/overlays bhi audit mein ☐

Screen sirf adhi kahani hai — is liye audit ko do aur muddaton(par) barhaya gaya:

- **Dark theme** (`node tools/audit_layout.js 1440 900 --dark`) — poori app `[data-theme="dark"]` mein chalakar wahi 6 checks. Dark mode mein **16 masle** mile, sab theek:
- `.btn.ok` (safed text on bright green) sirf **2.28:1** → naye tokens `--on-ok` / `--on-err` / `--on-brand` : light theme mein safed ink, dark theme mein gehra ink (green par 6.5:1, red par 5.3:1).
- dark `--muted-2` (#64748b) 3.75:1 → **#8b98ad** (6.1:1), `--muted` → **#a3b1c6**.
- **Overlays / modals** (`--overlays`) — 8 modal & offcanvas (export/share dialog, **day closing report**, **print preview**, user form (rules), item detail, PO form, count sheet, voucher) khol kar alag se nape jate hain (screen ke andar chhupe masle bhi pakde jayein). Yahan offcanvas/modal ka × **close button 30px** tha → ab **34px** (touch-friendly).

light 1440×900 → 0 light 390×844 → 0 dark 1440×900 → 0
overlays 1440×900 → 0 overlays 390×844 → 0

17. Sheets / DB schema — ab sahi chalta hai (silent data loss khatam) ☐

Masla: `DB.insert('Sales', { prevBalance: 2500 })` — agar `prevBalance` `SCHEMA.Sales` mein declared na ho to **DB.insert use chupke se drop kar deta hai** (DB.gs sirf declared columns likhta hai). Koi error nahi, koi warning nahi — bas data ghaayab.

Naya gate `node tools/audit_schema.js` har `DB.insert` / `insertMany` / `update` call ke object ke har key ko `SCHEMA` columns se match karta hai. Pehle run mein **7 asli data-loss bugs** mile:

Sheet	Field	Kahan	Status
Notifications	locationId	Notifications.gs ×2	column add kiya (branch-wise alerts)
CountSheets	updatedAt , postedBy	Warehouse.gs ×3	columns add (count sheet audit trail)
Items	active (ghalat naam)	PriceImport.gs	code theek → <code>status: 'ACTIVE'</code>
Items	supplierId (ghalat naam)	PriceImport.gs	code theek → <code>primarySupplierId</code>

Schema ab UI se bhi repair hota hai (pehle sirf Apps Script editor se ***Setup*** chalana padta tha):

- Naye backend functions: `Setup.schemaHealth()` (kaunsi sheet ghaayab, kaunse columns missing, row counts) aur `Setup.repair()` (ghaayab sheets banayein + missing columns append — **data delete nahi hota**). Routes: `setup.health`, `setup.repair`.
- **Settings ► Data & tools ► Maintenance** mein naya panel  **Sheets / DB schema**: Sheets / Columns / Rows / Schema KPIs, missing columns ki table, aur  **Repair / upgrade sheets** button +  Recheck.
- `Setup.createSheet` ab **khali sheet** (0 columns) par bhi headers likhta hai (pehle header row ghalat jagah ban rahi thi).
- About tab ke hardcoded “Version 2.0 / 20 files” hata kar **asli** version + sheets/rows counts (`system.health` se).

Tests: 10 naye assertions — schema round-trip (likha gaya field sheet mein wapas milta hai), health check, repair (naya column add, ghaayab sheet dobara bane, header blank hone par pakad), aur “repair se koi row delete nahi hoti”.

18. Sheets cells — object/array/boolean ab sahi save hote hain

Doosra chhupa hua data-loss: `DB.insert / update` object ya array ko seedha sheet cell mein likhta tha → cell mein `[object Object]` (ya `[object Object],[object Object]`) save ho jata tha. Koi error nahi, bas data ghaayab. Boolean `TRUE/FALSE` ban kar `split` /comparison todte the.

Probe se saboot (pehle):

```
rows    → "[object Object],[object Object]"    errors → "[object Object]"
trackSerial → "TRUE"      hasExpiry → "FALSE"
```

Fix — `DB._cell()` (DB.gs) har likhi jane wali value ko cell ke layak banata hai: `object/array → JSON string` · `boolean → 'true'/'false'` · `Date → ISO` · `null/undefined → ''`. Lagoo: `DB.insert` · `DB.insertMany` · `DB.update` · `DB.setRowValues`. Sath hi read-side helper `DB.json(cell, fallback)` — JSON column wapas object/array de deta hai.

Probe ke baad:

```
rows    → "[{\"a\":1},{\"a\":2}]"    errors → "{\"row1\":\"bad\"}"
trackSerial → "true"      hasExpiry → "false"    null → ""
```

Regression guard: naye columns JSON likhte hain, par kuch purane columns aaj bhi `split(',')` se parhe jate hain (`Users.locationIds`, `extraPermissions`, ...). Is liye `audit_schema.js` ab ye bhi check karta hai: * **JSON vs split mismatch** — comma-list column mein array bheja gaya (`.join(',')` ke baghair) → fail * **NumberSeries coverage** — `DB.nextNumber('X')` ka prefix defined na ho to number `DOC-00001` ban jata hai; ab 13/13 entities covered hain * dono checks ka self-test kiya gaya (ghalat entity + array write daal kar dekha → dono pakde)

Tests: 6 naye assertions — array/object round-trip, Date → ISO, boolean → `'true'/'false'`, `null` → `''`, aur number-series coverage.

19. Asli product master — 422 products CSV se DB mein

"add these all products (attached product list2.csv) and details in the db, seed them"

Ab demo/heuristic catalogue nahi — **aapki apni file** hi system ka product master hai. Har CSV column kahin na kahin map hota hai; **koi bhi column raaste mein nahi girta**.

Pipeline (ek hi command, dobara chalane layak)

```
product list2.csv
└─ python3 tools/import_products_csv.py    → tools/products.json    (+ data-
quality report)
    └─ python3 tools/gen_seed.py           → apps-script/Seed_Products.gs
                                          → demo/products.js
                                          → python3 tools/build_demo.py
```

Apps Script editor mein: `seedRealProducts()` — masters + items + stock, sab kuch.

Ya bina editor khole — app se: Settings ► Data & tools ►  Real catalog

Button	Kya karta hai
↓ Seed 422 products	Asli import — items + masters + stock rows
👁 Preview (safe)	Kuch write nahi karta — sirf dikhata hai ke kitne naye banenge / update honge
🔍 Audit CSV ↔ DB	Import ke baad milan check: kitne products ghaayab hain, kitne rate farq hain

Bug fix: is screen ka *Preview* button pehle `dryRun` bhejta tha magar seeder usay nahi samajhta tha → **Preview dabane par asal mein seed ho jata tha** (`stock: 0` ke sath). Ab `seedRealProducts({dryRun:true})` sirf hisaab lagata hai — **koi write nahi** (naye tests: dry run ke baad Items / Stock / Categories / Suppliers / Brands ke row counts bilkul wahi rehte hain).

Column map (CSV → sheet)

CSV column	Kahan gaya
Product Code	<code>Items.code</code>
Product Name	<code>Items.name</code>
Line Item	<code>Items.lineItem</code> (naya column, level-1) + <code>Categories</code> root node

CSV column	Kahan gaya
Category	<code>Items.category</code> + <code>Categories</code> child node
Sub Category	<code>Items.subCategory</code> (level-3)
Brand	<code>Items.brand</code> + Brands master (auto-create)
Model	<code>Items.model</code>
Made in	<code>Items.origin</code>
Conversion Factor	<code>Items.conversionFactor</code> (0 /blank → 1 , kabhi 0 nahi)
Size	<code>Items.size</code> ("Default" → khali)
Supplier	<code>Items.primarySupplierId</code> + Suppliers master (auto-create)
Cost Price	<code>Items.costPrice</code>
Wholesale Price	<code>Items.wholesalePrice</code>
Retail Price	<code>Items.retailPrice</code>
Barcode	<code>Items.barcode</code> (khali ho to product code — scanner foran chale)
Product Group / Color / Cost Type / Purchase Type	<code>Items.customFields</code> (JSON: <code>cf_productGroup</code> , <code>cf_color</code> , <code>cf_costType</code> , <code>cf_purchaseType</code>)
Profit(%) / Difference	derived (<code>Difference == WP - CP</code>) — JSON mein mehfooz

Jo ab banta hai

Cheez	Count
Products	422
Line items	10
Categories	95
Sub-categories	78
Brands (auto-create)	74
Suppliers (auto-create)	7
Made in (origins)	8
Real barcodes	87 (baqi 335 product code par fallback)
Stock rows	$422 \times 3 \text{ branches} = \mathbf{1,266}$

Idempotent — jitni baar chalayein

`seedRealProducts()` dobara chalane par **0 naye items, 0 naye stock rows, 0 naye suppliers/brands/categories** — sirf existing rows update hoti hain (422 updated). Koi row delete nahi hoti.

`seedRealProducts({prices:false})` se rates chhoday baghair sirf naye products add ho sakte hain.
`auditRealProducts()` CSV aur DB ka milan check karta hai (dry run).

Naya column: `Items.lineItem`

Aapke system ki 3-level taxonomy (`Line Item` → `Category` → `Sub Category`) ko sahi se rakhne ke liye `Items` sheet mein `lineItem` column add hua hai. * `Items.list({lineItem})` filter · `Items.facets().lineItems` dropdown * Items screen: **Line item** column + filter + item form mein field * **Existing spreadsheet par ek baar Settings ► Data & tools ► Maintenance → ☐ Repair / upgrade sheets chalayein** — column append ho jayega, data safe.

Bug fix: item form detail uda raha tha

`Items.FIELDS` mein sirf purane 35 column the. Items screen se **Save** karte hi `size` , `origin` , `conversionFactor` , `primarySupplierId` , `customFields` , `variants` , `lineItem` **khamoshi se delete** ho jate the. Ab `Items.FIELDS` पूरी `SCHEMA.Items` list cover karta hai (test: har FIELDS entry declared column hai).

Data quality — aapka faisla darkaar ⚠

Import **bilkul faithful** hai: file jaisi thi waisi hi chali gayi, hamari taraf se **koi qeemat nahi badli**.
Report: `release/product-import-report.html`

Finding	Count	Kya karein
☐ Wholesale price > Retail price	133	File mein hi aisa hai (ya columns swap hain). Faisla karne ke baad <code>python3 tools/import_products_csv.py --price-mode swap</code> (ya <code>clamp</code>) → <code>python3 tools/gen_seed.py</code>
● Cost aur Retail dono 0	12	Items screen ya price import se rates bharein
● Supplier khali	86	<code>primarySupplierId</code> khali hai — kharid/reorder ke waqt supplier chunein
☐ Barcode khali (code use hua)	335	Asli barcode milne par Items ► Barcode update karein

Naye tests

`tools/test_logic.js` mein **57 naye assertions** (532 → **589**): catalogue ↔ seed ↔ sheet milan, prices exact, lineItem/category/subCategory/ brand/model/origin/size/conversionFactor/barcode loss-check, custom fields JSON (kabhi `[object Object]` nahi), masters duplicate-free, `primarySupplierId` dangling nahi, har item × branch stock row, `[object Object]` /raw-boolean cell nahi, **idempotency (doosra run)**, `prices:false` izzat, `auditRealProducts()` 0 missing / 0 price-drift, `Items.FIELDS` coverage, `lineItem` filter, aur **dry-run** (Preview button kuch write na kare).

Test gates (v2.5.2)

Gate	Result
<code>bash tools/check.sh</code> (Apps Script + template syntax)	29 blocks / 0 invalid
<code>node tools/audit_schema.js</code> (Sheets/DB schema health)	67 sheets · 825 columns · 0 problems
<code>node tools/test_logic.js</code> (backend logic)	589 / 0
<code>node tools/smoke.js</code> (jsdom UI)	308 / 0
<code>node tools/audit_tabs.js</code>	483 / 0
<code>node tools/audit_align.js</code>	36 / 0
<code>node tools/verify_punchlist.js</code> (16 requirements)	16/16 · 117 checks / 0
<code>node tools/audit_layout.js 1440 900</code> (real browser)	0 problems
<code>node tools/audit_layout.js 390 844</code> (mobile)	0 problems
<code>node tools/audit_layout.js 1440 900 --dark</code>	0 problems
<code>node tools/audit_layout.js --overlays</code> (8 modals × desktop/mobile)	0 problems

`bash tools/verify.sh` sab ek sath chalata hai.

☐ Haseeb Autos — Google Apps Script se Deploy Karne Ka Mukammal Guide

Roman Urdu + English mix. Zero se le kar live app tak — har step, har command, har error ka hal. Poora app **Google Apps Script + Google Sheets** par chalta hai: koi server nahi, koi hosting fee nahi, sirf aap ka Google account.

📌 30 second mein summary

Step	Kaam	Time
1	Google account + naya Apps Script project	2 min
2	Code upload (clasp ya copy/paste)	3 min
3	<code>setupAll()</code> run → Sheets database ban jata hai	2 min
4	Deploy ► New deployment ► Web app	2 min
5	Pehli login (<code>owner</code> / <code>admin123</code>) + password badlein	2 min
6	AI key + daily jobs (optional)	3 min

Kul: ~15 minute. Neechay har step detail mein hai.

☐ Pehlay ye cheezein ready rakhein

- ☐ **Google account** (Gmail) — personal ya Workspace, dono chalte hain
- ☐ **Google Drive** ki permission (Sheets banane ke liye)
- ☐ (Optional) **Node.js 18+** — sirf clasp wale tareeqe ke liye
- ☐ Is folder ka ZIP (ya clone) apne computer par

Note: App aap ke Google account ki **quota limits** use karta hai (20,000 UrlFetch calls/day, 6 min execution time). Retail shop ke liye ye bohat zyada hai.

A Option A — clasp se deploy (recommended, ek command)

clasp Google ka official command-line tool hai — saari files ek sath upload ho jati hain, naam/ghalti ka koi chance nahi.

A1. clasp install + login

```
# 1. Node.js install hai? check karein
node -v          # v18 ya upar hona chahiye

# 2. clasp install
npm install -g @google/clasp

# 3. Google login (browser khulega → Allow karein)
clasp login
```

Agar **clasp login** mein error aaye to: script.google.com/home/usersettings → **Apps Script API** ko **On** karein.

A2. Naya project banayein

```
cd haseeb-autos/apps-script

# Naya project (Script + bound Spreadsheet ek sath)
clasp create --title "Haseeb Autos ERP" --type standalone
```

Ya **existing project** ke sath:

1. script.google.com → **New project**
2. **Project Settings (⚙)** → **Script ID** copy karein
3. **apps-script/.clasp.json** banayein:

```
{ "scriptId": "1AbC...xyz", "rootDir": "apps-script" }
```

A3. Push (code upload)

```
clasp push          # sab .gs + .html files upload
clasp open          # Apps Script editor khul jayega (browser)
```

Pehli dafa `.claspignore` ensure karein: sirf `.gs` aur `.html` jayen, `appsscript.json` bhi.

A4. Database banana (sab se zaroori step)

Apps Script editor mein:

Function select: `setupAll` → Run ▶

1. **Authorization required** dialog aayega → **Review permissions** → apna account → **Advanced ▶ Allow**
2. Execution log mein ye nazar aayega:

```
{ "spreadsheetId": "1XyZ...", "url": "https://docs.google.com/spreadsheets/d/1XyZ.../edit",  
  "jobs": { "installed": [ {"handler": "triggerDailyReorder"},  
    {"handler": "triggerDailyAlerts"} ] },  
  "message": "Setup complete. Ab Deploy > New deployment > Web app karein." }
```

□ Ab aap ki **"Haseeb Autos - ERP Database"** spreadsheet ban chuki hai: 30+ sheets, 3 branches (SDQ / MCH / RYK), 9 users, 101 catalogue items, print templates.

Spreadsheet ka link save kar lein — yahi aap ka **database** hai.

A5. Web app deploy

Deploy ▶ New deployment ▶ (⚙ gear) Web app

Field	Value
Description	Haseeb Autos v2.1.0
Execute as	Me (apna account)
Who has access	Anyone (ya "Anyone with Google account" — zyada secure)

Deploy dabayein → pehli dafa dobara **Authorize access** karein → **Web app URL** milega:

`https://script.google.com/macros/s/AKfycb.../exec`

□ **Yehi aap ke app ka address hai.** Mobile/desktop browser mein kholein, ya QR bana kar staff ko bhej dein (App ke **Settings ▶ Business** mein QR print bhi ho sakta hai).

B Option B — bina Node ke (copy/paste)

Agar Node install nahi kar sakte to ye tareeqa use karein. ~10 minute lagte hain.

B1. Project + files

- 1. script.google.com → New project
- 2. Project ka naam: **Haseeb Autos ERP**
- 3. Har file ke liye: + (Plus) ▶ Script ya ▶ HTML - `.gs` files → Script - `.html` files → HTML (naam **bina extension** ke — masalan `Index` , `Styles` , `App_Core`)
- 4. Is folder ki `apps-script/` se content copy kar ke paste karein

Server files (Script) — isi order mein banayein

#	File	Kya karta hai
1	Utils	Helpers, CONFIG, formatting, <code>U.*</code>
2	DB	Sheets read/write layer
3	Auth	Login, sessions, RBAC
4	Audit	Audit log
5	Schema	Har sheet ke columns (single source of truth)
6	Config	Settings registry (<code>Config.defs()</code>)
7	Setup	Sheets banana + seed data
8	Items	Product master
9	Parties	Customers / Suppliers / Ledger
10	Inventory	Stock, transfers, adjustments
11	Sales	POS billing, returns, holds
12	Purchase	PO, GRN, purchase return
13	Payments	Cash in/out, expenses
14	Reports	Dashboards, P&L, exports
15	Warehouse	Bins, put-away, count sheets
16	Notifications	Alert engine
17	Commissions	Salesman commission
18	Barcode	Barcode/QR label data
19	OfflineSync	Offline pull/sync
20	AI	AI agent (Gemini/OpenAI/OpenRouter)

#	File	Kya karta hai
21	Lang	Translations (EN/Roman/Urdu)
22	Reorder	Auto reorder engine (v2.1)
23	Triggers	Scheduled jobs (v2.1)
24	Seed_Products	71 asli Flamingo products
25	Code	doGet(), api() router

Client files (HTML) — bina .html extension ke

Index, Styles, App_Core, App_Barcode, App_QR, App_UI2, App_POS, App_Screens, App_Screens2, App_POS2, App_Masters, App_Inventory2, App_Config, App_Dashboards, App_Lang, App_Reorder, App_AI, App_Boot

⚠ **Naam bilkul waise hi rakhein** (case-sensitive). Ghalti se app blank screen dega.

B2. Apps Script manifest

Project Settings ⚙ → Show "appscript.json" → ye content daalein:

```
{
  "timeZone": "Asia/Karachi",
  "dependencies": {},
  "exceptionLogging": "STACKDRIVER",
  "runtimeVersion": "V8",
  "webapp": { "executeAs": "USER_DEPLOYING", "access": "ANYONE_ANONYMOUS" }
}
```

B3. Ab wahi A4 + A5

setupAll() run karein → authorize → Deploy ▶ New deployment ▶ Web app.

⚙ Deploy ke baad ki zaroori settings

1. Pehli login (aur foran password badlein)

Role	Username	Password
Owner	owner	admin123

Role	Username	Password
Manager	manager	manager123
Salesman	sales1	sales123
Purchase	purchase	purchase123
Warehouse	warehouse	ware123
Accountant	accountant	acc123
Cashier	cashier	cash123
Delivery	delivery	del123

☐ **Sab se pehla kaam:** `Users & Security` → har user ka password change karein.

2. Business profile

`Settings ▶ Business ▶ Identity` — naam, phone, address, NTN, currency. Ye sab **Sheets** mein save hota hai (kuch bhi hardcoded nahi).

3. Catalogue import (apna purana stock)

`Items` screen → **↑ Import** (CSV) ya `Settings ▶ Data & tools ▶ Import`. Format: `code, name, category, brand, cost, retail, wholesale, stock`.

4. AI Assistant (optional, 2 minute)

1. aistudio.google.com/apikey → **Create API key**
2. Apps Script editor: **Project Settings** ⚙ → **Script Properties** → **Add** `Name: AI_API_KEY Value: AIza...`
3. App mein: `Settings ▶ Automation ▶ AI assistant` → Provider `GEMINI`, Enable ☑
4. `aiCanWrite = false` rakhein jab tak aap AI ko direct write nahi karwana chahte (default: AI sirf **draft** banata hai, aap confirm karte hain)

5. Daily jobs / triggers (v2.1)

Editor mein (ya `Settings ▶ Automation ▶ Scheduled jobs ▶ ☐ Install jobs` se):

Function: `installTriggers` → Run ▶ → Authorize

Setting	Default	Matlab
<code>job.dailyReorder</code>	<code>true</code>	Roz subah <code>Reorder.run()</code> — suggestions + notification
<code>job.dailyHour</code>	<code>7</code>	Kab chalay (Asia/Karachi)

Setting	Default	Matlab
<code>job.dailyAutoPO</code>	<code>false</code>	Suggestions se khud PO banayein
<code>job.dailyAlerts</code>	<code>true</code>	Low stock / udhaar / pending PO alerts
<code>job.alertsHour</code>	<code>6</code>	Alerts ka waqt

Editor se test ke liye: `runReorderNow()` ya `triggerStatus()` chala kar dekhein.

□ Mobile PWA (app ki tarah install)

Android (Chrome): 1. Web app URL kholein → ⋮ menu → **Install app / Add to Home screen** 2. Icon home screen par aa jayega — offline bhi khulta hai (catalog cache)

iPhone (Safari): Share ► **Add to Home Screen**

Apps Script ke `/exec` URL par app **khud** apna manifest inject karta hai (data-URI), is liye install prompt bina kisi extra file ke aa jata hai. Asli `sw.js` (offline shell) sirf static hosting par chalta hai — `DEPLOYMENT.md` ► `Option C`.

□ Updates kaise karein (naye features)

```
# clasp wale
clasp push
```

Phir editor mein:

Function	Kab chalayein
<code>repairSchema</code>	Naye sheets/columns add karne hain (purana data safe rehta hai)
<code>seedAll</code>	Catalog lists, warehouses, templates dobara seed karne hain
<code>seedTranslations</code>	Urdu/Roman dictionary refresh karni ho (v2.1)
<code>reinstallTriggers</code>	Jobs dobara install karne hain

Deployment update: `Deploy` ► `Manage deployments` ► ⓘ `Edit` ► `Version: New version`.

⚠ Zaroori: naya version banayein, warna users ko purana code dikhega (Apps Script caching).

Version badhane ka sahi tareeqa: `apps-script/Utils.gs` mein `CONFIG.VERSION` badlein → demo build aur service-worker cache khud update ho jata hai.

Deploy se pehle verification (developer)

```
bash tools/check.sh          # 23 .gs + HTML script blocks syntax
node tools/test_logic.js     # 97 business-logic tests (real backend, mock Sheets)
node tools/smoke.js          # 109 UI checks – har screen, har tab, deep links
node tools/ui_audit.js       # UI defect scan (labels/titles/blank content)
node tools/test_qr.js && python3 tools/verify_qr.py
```

Sab green hon → `clasp push`.

SOS Troubleshooting

Masla	Wajah	Hal
"Authorization required" / "This app isn't verified"	Google ka standard warning	Advanced ► Go to Haseeb Autos (unsafe) ► Allow
<code>Setup.setupAll</code> mein "Sheet not found"	Spreadsheet link toota	Editor: <code>Setup.attach("<SPREADSHEET_ID>")</code> phir <code>setupAll()</code>
Blank white screen	HTML file ka naam ghalat (extension samet)	File names check karein: <code>Index</code> , <code>Styles</code> , ... (<code>.html</code> ke baghair)
<code>ScriptApp</code> / trigger install error	Trigger ke liye authorization pending	<code>installTriggers()</code> editor se run karein → Allow
Job chal raha hai par notification nahi	<code>job.*</code> settings off, ya item hi reorder par nahi	Settings ► Automation ► Scheduled jobs + Reorder screen check karein
"Service invoked too many times"	Sheets quota (writes/min)	Thora wait karein; bulk import chhote batches mein karein
Execution timeout (6 min)	Bohat bada import/export	Import ko 500-row batches mein split karein
Login ke baad "Session expire"	Cache TTL (8 ghante)	Dobara login; <code>sec.sessionTimeout</code> setting se badal sakte hain
AI jawab nahi de raha	<code>AI_API_KEY</code> missing ya quota	Script Properties check; Settings ► AI provider/model
Urdu/Roman switch nahi ho raha	Dictionary seed nahi hui	<code>seedTranslations()</code> run karein; phir <code>#EN</code> button (dbl-click = editor)

Masla	Wajah	Hal
Barcode scanner kaam nahi (mobile)	Camera permission denied	Browser settings → Site permissions → Camera: Allow (HTTPS lazmi)
Naye code ke baad purana UI	Deployment version update nahi hua	Manage deployments ► Edit ► New version
Slow first load	Apps Script cold start (2-4 s)	Normal hai; baar baar use par fast ho jata hai

☐ Daily use checklist

Subah (shop kholte waqt) 1. **Shop Open/Close** → opening cash daalein → session start 2. ☐ **Alerts** check karein (low stock, udhaar, pending PO) 3. ☐ **Auto Reorder** → suggestions dekhein → zaroori lines select → **Create POs**

Din mein 4. **POS** → cards se item tap karein / barcode scan → qty → Cash/Udhaar 5. Udhaar → **Parties** → customer ledger

Shaam (shop band karte waqt) 6. **Reports** → day sale, cash, P&L 7. **Shop Open/Close** → counting → variance → close 8. Haftay mein ek dafa: **Settings ► Data ► Backup** (JSON export) Drive mein save karein

Version: 2.1.0 · **Stack:** Google Apps Script (V8) + Google Sheets · **License:** Aap ki property

Deployment Guide — Haseeb Autos ERP & POS

Ye guide 3 tareeqay deta hai. **Option A (clasp)** sab se fast hai, **Option B (copy/paste)** sab se simple.

Pehlay ye check list

- [] Google account (jis se system chalana hai — ideally business ka apna account)
- [] Node.js 18+ (agar clasp use karna hai) — nodejs.org
- [] Gemini API key (AI ke liye) — aistudio.google.com/apikey (optional)
- [] 15–20 minute

Option A — clasp (recommended, one command)

1. clasp install + login

```
npm install -g @google/clasp
clasp login          # browser khulega → Google account allow karein
clasp enable apps-script-api # sirf ek dafa (https://script.google.com/home/usersettings)
```

2. Apps Script project banayein aur `.clasp.json` mein ID daalein

```
# Naya project (script + bound spreadsheet ek saath)
clasp create --title "Haseeb Autos ERP" --type standalone
# ✓ .clasp.json automatically ban jata hai

# Ya existing project ke sath:
# script.google.com → project kholein → Project Settings → Script ID copy
# phir .clasp.json mein: { "scriptId": "1AbC...xyz", "rootDir": "apps-script" }
```

`tools/.clasp.json.template` ko copy kar ke root mein `.clasp.json` bana dein.

3. Push + setup

```
cd haseeb-autos
bash tools/deploy.sh          # → clasp push

# ab Apps Script editor mein:
#   function select: setupAll → Run ► → authorize → Logs dekhain
clasp open                    # editor kholta hai
```

Editor mein **setupAll** run karne ke baad Logs mein aayega:

```
{ spreadsheetId: '1xYz...', url: 'https://docs.google.com/spreadsheets/d/1xYz.../
edit',
  message: 'Setup complete. Ab Deploy > New deployment > Web app karein.' }
```

4. Deploy (web app)

```
clasp deploy --description "Haseeb Autos v1.0"
```

Ya editor se: **Deploy ► New deployment ► ⚙ Web app** - Execute as: **Me** (aapka Google account) - Who has access: **Anyone** (login app ke andar hai, is liye safe hai) - **Deploy** → URL copy karein → **Ctrl+V** browser mein ☐

Option B — copy/paste (bina Node ke)

1. script.google.com → + **New project**
2. Project name: **Haseeb Autos ERP**
3. Har **.gs** file ke liye: + ► **Script** → file name (bina **.gs**) type karein → content paste → ☐ Save - **Schema**, **Utils**, **DB**, **Setup**, **Auth**, **Audit**, **Items**, **Inventory**, **Sales**, **Purchase**, **Parties**, **Payments**, **Reports**, **AI**, **Barcode**, **OfflineSync**, **Code**
4. Har **.html** file ke liye: + ► **HTML** → name (**Index**, **Styles**, **App_Core**, **App_Barcode**, **App_POS**, **App_Screens**, **App_Screens2**, **App_AI**, **App_Boot**) → content paste → Save
5. ⚙ **Project Settings** → tick **Show "appsscript.json"** → wapas editor → **appsscript.json** ka content replace karein
6. Function dropdown mein **setupAll** select → **Run ►** → **Review permissions** → **Advanced** → **Allow**
7. Logs mein spreadsheet URL aayega
8. **Deploy ► New deployment ► Web app** (Execute as: Me · Access: Anyone) → Deploy

Option C — static PWA hosting (advanced, real service worker)

Agar aap chahte hain ke app **real installable PWA** ho (offline app shell, add-to-home-screen prompt):

1. `demo/index.html` (ya apps-script templates se build) ko **Netlify / Vercel / GitHub Pages / Firebase Hosting** par host karein.
2. Apps Script project ko **Web app** ki tarah deploy karein (Mode A/B ki tarah) aur uska `/exec` URL lein.
3. Frontend mein `window.API_URL` set karein: - `Index.html` mein: `<script>window.API_URL = 'https://script.google.com/macros/s/AKfy.../exec';</script>` - App automatically **JSONP** transport use karega (CORS problem nahi aati).
4. `python3 tools/build_demo.py` chalayein — yeh `demo/manifest.webmanifest`, `demo/sw.js` aur `demo/icon-192.png` / `icon-512.png` khud bana deta hai (source icons: `tools/icon-192.png`, `tools/icon-512.png`). Poori `demo/` folder (index.html + manifest + sw + icons) static host par upload karein.
5. Naye build par service-worker cache refresh karne ke liye `tools/build_demo.py` ke andar `CACHE = 'haseeb-autos-v...'` version badhayein — purana cache auto-clear hoga.

Note: Apps Script ke `/exec` scope me alag file serve nahi ho sakti, is liye wahan app apne aap **inline data-URI manifest** use karta hai (install prompt phir bhi aata hai). Asli `sw.js` sirf static hosting par register hota hai (`window.SERVE_MODE` check).

Faida: poora app offline (app shell cache), install prompt, faster loading. **Nuksan:** do jagah deploy karna parhta hai.

□ AI Assistant setup (2 minute)

Gemini (recommended, free tier kaafi hai)

1. aistudio.google.com/apikey → **Create API key**
2. Apps Script editor → ⚙ **Project Settings** → **Script Properties** → **Add script property** - Name: `AI_API_KEY` · Value: `AIza...`
3. App mein **Settings** ▶ **AI Assistant** → provider `GEMINI`, model `gemini-2.0-flash` (default hai)
4. Test: POS ke upar □ button → "Aaj ki sale kitni hui?"

Ya editor mein directly run karein:

```
setAIKey("AIzaSy...", "GEMINI", "gemini-2.0-flash")
```

OpenAI / OpenRouter

Settings ► AI Assistant ► Configure provider: - `OPENAI` + `gpt-4o-mini` (key OpenAI ki) - `OPENROUTER` + `anthropic/claude-3.5-sonnet` etc.

AI write mode

Default: **draft only** (AI sirf draft banata hai, kuch post nahi hota). Agar aap chahte hain ke AI se PO/sale draft ban kar seedha save ho: Settings ► `aiCanWrite = true` (pehlay kuch din draft mode mein chalayein).

□ Pehli login ke baad kya karein

1. `owner / admin123` se login → **Settings ► Users** → har user ka password change karein (ya username click kar ke **Change password**)
2. Har user ko uski **branch access** aur **role** assign karein
3. **Settings ► Configuration ► Business** — naam, phone, address, NTN, tax %, currency
4. **Settings ► POS** — cards per row, default view, stock badge, quantity pad, quick cash
5. **Settings ► Lists & catalogs** — apni Categories / Brands / Units / Tax codes
6. **Settings ► Menu builder** — sidebar ke items apni marzi se (label, icon, order, permission)
7. **Settings ► Print templates** — receipt/label JSON (barcode/QR, paper size, footer)
8. **Settings ► Data ► Real catalog** — "**Seed 71 products**" (Flamingo catalogue, list.pdf se)
9. **Items** screen se baqi stock import karein (CSV paste — Import button)
10. **Customers** mein purana udhaar (opening balance) daal dein
11. **Inventory ► Warehouse** — apne racks/bins banaayein (ya `Setup.seedWarehouses()` run karein)
12. Mobile par URL khol kar **Add to Home Screen** (PWA install)

📁 Data import (purana stock / customers)

Items: Items screen ► **Import** → Excel/CSV se copy kar ke paste (tab-separated bhi chalta hai)

code	name	category	brand	costPrice	retailPrice	wholesalePrice	barcode
HLG-001	LED Headlight	Lights	Philips	850	1200	1050	HLG-001

- `code` existing ho ga to **update**, nahi to **naya item** banay ga.

Customers: Parties ► Add Customer (ya `Setup.seedDemoItems()` ki tarah bulk loader likhwa lein).

Opening stock: Inventory ► Adjustment → reason `OPENING` → items add kar ke **Save & Post**.

v2.1 upgrade (existing install par)

```
# clasp mode
clasp push
# phir Apps Script editor mein:
#   function: repairSchema      → Run ►   # Translations / SavedViews + lineBase
column (data safe)
#   function: seedAll          → Run ►   # + seedTranslations() (EN/Roman/Urdu
dictionary)
#   function: setupAll         →   sirf naye project par
```

Naye Settings keys (Settings ► ... se editable): `discountBeforeTax`, `taxInclusive`, `costingMethod` (AVG/FIFO), `maxDiscountPct`, aur **Automation ► Purchase reorder rules** group (`autoReorder*` keys).

v2 deployment — naye sheets aur data

v2 mein **14 naye sheets** judti hain: `Categories`, `Brands`, `Units`, `TaxCodes`, `Warehouses`, `Bins`, `BinStock`, `StockLots`, `Bundles`, `PriceLists`, `CustomFields`, `MenuConfig`, `PrintTemplates`, `SavedViews` — aur `Items` / `Stock` / `Customers` / `Suppliers` mein naye columns.

Agar aap v1 se upgrade kar rahe hain:

```
# clasp mode
bash tools/deploy.sh
# phir Apps Script editor mein:
#   function: repairSchema → Run ►   # naye sheets + columns add (data safe)
#   function: seedAll     → Run ►   # catalog lists + warehouses + templates
```

`repairSchema()` purana data delete nahi karta — sirf missing sheets/columns add karta hai.

71 asli products (Flamingo catalogue): - Apps Script: `Setup.seedProducts()` → `seedRealProducts({ stock: 20 })` - Ya UI: **Settings ► Data & tools ► Real catalog ► Seed 71 products** - Har item ke sath 3 branches ke liye stock row ban jati hai

Deployment se pehle verification (developer)

```
bash tools/check.sh          # 23 .gs files + 19 script blocks syntax
node tools/test_logic.js     # 78 business-logic tests (costing, tax, ledger,
reorder, i18n)
node tools/smoke.js          # 100 UI checks – har screen + har tab + deep links + QR
node tools/test_qr.js && python3 tools/verify_qr.py
# tools/mock_gs.js → in-memory Apps Script runtime (SpreadsheetApp/CacheService
fake)
#                             jo test_logic.js real .gs files ke saath use karta hai
```

Sab green hone ke baad hi `deploy.sh` chalayein.

Troubleshooting

Masla	Hal
<code>SPREADSHEET_ID</code> not set	<code>setupAll()</code> run karein (function dropdown se)
Login ke baad blank screen	Browser console (F12) dekhein; aksar <code>AI_API_KEY</code> missing hone pe AI hi error deta hai, baqi app chalay ga
<code>Session expire</code> ho gaya	Dobara login (12h baad hota hai)
Stock update nahi ho raha	Cache 5 min ka hai — Refresh dabayein; phir bhi na ho to Settings ► Data ► Row counts
AI jawab nahi deta	Settings ► AI Assistant ► key check karein; Gemini quota
Barcode scan nahi ho raha	Scanner Code39 mode par ho (default hota hai); item ka barcode khali na ho
Print receipt khali aata hai	Popup blocker allow karein
Bahut slow ho gaya	Sheet bari ho gayi — purana data purge karein (Settings ► Data)
Naye columns chahiye	<code>Schema.gs</code> mein column add karein → <code>repairSchema()</code> run karein
Naye fields chahiye (bina code)	Settings ► Custom fields — Item/Customer/Supplier par apna field banaayein
Menu item chhupana hai	Settings ► Menu builder → item edit → Active off
Settings screen khali	<code>config.defs()</code> route deploy hua ya nahi check karein; <code>repairSchema()</code> run karein
Warehouse/Bins nazar nahi	<code>Setup.seedWarehouses()</code> run karein (ya Settings ► Lists ► Warehouses se add)
QR nahi ban raha	

Masla	Hal
	Ab offline encoder hai — internet zaroori nahi; data 134 bytes se kam hona chahiye

□ Updates kaise karein (naye features aane par)

```
git pull          # agar repo use kar rahe hain
bash tools/check.sh # syntax verify
bash tools/deploy.sh
# Apps Script editor → Deploy ► Manage deployments → Edit → Version: New version
# (Naya deployment banane se URL change ho jata hai – Edit use karein taake URL same rahe)
```

Schema change ho to: editor mein **repairSchema** run karein (naye columns add ho jayen ge, data safe rahe ga).

□ Security best practices

- □ Pehlay din har default password badlein
- □ Spreadsheet ko **kisi ke sath publicly share na karein** — sirf Apps Script (Execute as: Me) access karta hai
- □ Owner account par 2FA on karein
- □ Har 1–2 hafte mein **Settings ► Data ► Backup to Drive**
- □ AI key kabhi bhi sheet mein na likhein (Script Properties mein rahe)
- □ Staff ko sirf zaroori role dein (Salesman ko **reports.financial** na dein)

□ Agla step

Kya chahiye next? 1. **WhatsApp/SMS** invoice bhejna (Twilio / Meta Cloud API) 2. **Supplier price lists** auto-import (email/CSV parsing) 3. **Customer loyalty** — points redeem at POS 4. **Multi-currency / purchase in USD** 5. **Firestore migration path** jab sheet bari ho jaye

Bas bata dein — agla feature bana deta hoon.

NEW Haseeb Autos — v2.5 Features Guide

Version **2.5.0** — accounting, real wallet payments, party-balance movement reports, a dedicated warehouse app, and a mobile order pad.

Roman Urdu mein likha gaya hai (har jagah EN/Urdu toggle bhi maujood hai).

1. **Accounting (proper double-entry)**

Screen: `Accounting` (permission: `reports.financial`)

Tab	Kya milta hai
Overview	Cash in hand, bank balance, receivables, payables, stock value, MTD profit, trial-balance health, recent vouchers
Transactions	Day book with date range + type filter, voucher detail, New voucher (double-entry, live Dr/ Cr difference)
Expenses	Expense KPIs, table, category breakdown, expense entry
Chart of accounts	Grouped COA (assets / liabilities / equity / income / expense), balances, new account, seed defaults
Banks	Bank book + reconciliation (CSV import → auto-match → clear/unclear)
Wallets	EasyPaiza / JazzCash requests with payer (from) + merchant (to) numbers
Reports	Trial balance, P&L (with COGS block), Balance sheet (with closing stock), Cash book, Account ledger

Conventions (backend + reports, ek hi tarteeb)

- Ledger balances **debit-positive**: `balance = opening + Σdebit - Σcredit`
- `normalSide`: **DR** for ASSET / EXPENSE, **CR** for LIABILITY / EQUITY / INCOME
- Trial balance: `bal > 0 → debit column`, `bal < 0 → credit column`
- Vouchers kabhi delete nahi hote — sirf **VOID** mark hote hain
- Auto-posting: `Config ▶ accounts.autoPost` (default **ON**) — sale, GRN, payment aur expense apna voucher khud banaate hain. Band karne par sirf haath ke vouchers aayenge.

Chart of accounts (default, editable)

`1000 Cash`, `1010–1030 cash counters`, `1100–1140 banks` (HBL, Meezan, JazzCash, EasyPaiza, Raast), `1200 Accounts receivable`, `1300 Inventory`, `2000 Accounts payable`, `2100 Tax payable`, `3000 Capital`, `3100 Retained earnings`, `4000/4100 Sales` (retail/wholesale), `4200 Other income`, `5000 Purchases / COGS`, `5100 Freight`, `6000–6900 expense heads`.

Settings: **Settings ▶ Trade & Accounts ▶ Chart of accounts** (har code tabdeel kar sakte hain) aur **Accounts** (`accounts.autoPost` , `accounts.showOnDashboard` , `accounts.fiscalCloseLock`).

Backfill

Purane data ke liye **Backfill vouchers** button (Overview tab) — sale / GRN / payment / expense se vouchers bana deta hai.

2. **Wallet gateway — EasyPaisha & JazzCash (asli API)**

Manual entry ab bhi kaam karti hai, lekin ab **asli gateway** se payment request bhi bhej sakte hain.

Setup

Settings ▶ Trade & Accounts ▶ Wallets

Setting	EasyPaisha	JazzCash
Mode	<code>OFF</code> / <code>SANDBOX</code> / <code>LIVE</code>	<code>OFF</code> / <code>SANDBOX</code> / <code>LIVE</code>
Merchant id	Store ID (numeric)	Merchant ID
Auth	API username + password	API password
Signing	Hash key (hosted checkout)	Integrity salt
Extra	EWP account number	Sub-merchant ID, Product ID
Endpoints	Sandbox / Live base URL (agar gateway path rotate kare to yahan edit)	same

Sandbox credentials EasyPaisha khud issue nahi karta — apne merchant rep se maangein.
JazzCash sandbox keys: <https://sandbox.jazzcash.com.pk>

Flow (POS par)

1. Payment modal mein method `EasyPaisha` / `JazzCash` chunein
2. Wallet panel mein **customer ka mobile (from)** aur **merchant wallet (to)** likhein
3. **Request bhejin** → gateway customer ke mobile par OTP / MPIN bhejta hai
4. Panel khud status check karta hai (`wallet.autoPollSeconds` , default 5s)
5. `PAID` hote hi payment row mein **txn ID** fill ho jata hai aur bill complete hota hai

Har request `WalletTxns` sheet mein record hoti hai (payer, receiver, amount, status, provider response) — **credentials kabhi log nahi hote**. Accounting ▶ Wallets tab se kisi bhi waqt status check / polling kar sakte hain.

Official integration notes (code mein bhi darj hain)

- **EasyPaisa REST v4:** `POST {base}/easypay-service/rest/v4/initiate-ma-transaction`, header `Credentials: base64(username:password)`, body `orderId, storeId, transactionAmount, transactionType:'MA', mobileAccountNo(03xxxxxxxxx), emailAddress, tokenExpiry`; inquiry `/inquire-transaction {orderId, storeId, accountNum}` → `transactionStatus: PAID | FAILED | PENDING | EXPIRED | REVERSED | BLOCKED`
- **JazzCash REST (v3.9):** `POST {base}/ApplicationAPI/API/Payment/DoTransaction` with `pp_*` fields; **amount paisa mein** (2500 Rs → `pp_Amount 250000`), payer number `ppmpf_1`, `pp_SecureHash = HMAC-SHA256(salt + '&' + sorted pp_* values)`; inquiry `/ApplicationAPI/API/Payment/Inquiry`, success `pp_ResponseCode = '000'`
- Fee / FED / net / settlement sab **Payment methods** engine se aate hain (koi duplicate formula nahi)

3. 🏢 Party balance movement — "pichhla baqaya + naya = kul baqaya"

Reports ► **Receivables / Payables / Party balances** ab 4 columns dikhaate hain:

Column	Formula
Prev (pichhla baqaya)	opening balance + ledger entries before the selected range
Naya / Purchase	range ke andar bana udhaar (customer) ya purchase (supplier)
Wasooli / Paid	range ke andar mili recovery / ki gayi payment
Closing	<code>Prev + Naya - Wasooli</code>

- Supplier ke liye payable **positive** rehta hai (`Parties.supplierPayable`)
- **to** date ke baad ki entries report se bahar rehti hain (as-of report)
- Naya report type: **Party balances (prev / new / closing)** — customer + supplier dono

Misaal: pichhla baqaya `2,500` + naya udhaar `1,000` = closing `3,500` .

4. 🏠 Warehouse app (count · audit · inventory)

Screen: **Warehouse** — ab 8 tabs:

Tab	Kaam
Overview	Warehouses, bins, units, value, low/out-of-stock, without-bin
Inventory	Branch / category / search filters, SKU value, low-stock switch, value-by-category chart, quick count

Tab	Kaam
Count sheets	Nayi sheet (warehouse / bin / category / search / zero / low scope) → blind entry → Post & adjust
Audit	Accuracy %, over/short, variance value, top variance, adjustments ledger (reverse bhi)
Movements	Poora stock ledger (in / out / balance / reference)
Bins & racks	Bin CRUD
Stock by bin	Bin-wise stock
Put-away / move	Item ko bin mein move karein

Count sheet ka sahi tareeqa

1. + **New count sheet** → scope chunein (system qty freeze ho jati hai — blind count)
2. Entry screen par code scan/type karein (Enter se us line par jump) aur counted qty likhein
3. Progress bar + live diff + variance value
4. **Post & adjust stock** → har farq ke liye `StockAdjustment` ban kar post hoti hai (stock + StockMoves + `lastCountedAt` stamp) — dobara post nahi ho sakti

Audit formula: `accuracy = matched lines / counted lines × 100`

5. **Order book + mobile order pad**

Screens: `Order Book` aur `Take Order (mobile)` (`pos.sell` permission)

- **Order Book:** list (status/customer search) + Pipeline (counts, open value, advance)
- Order detail: lines, status moves `DRAFT → CONFIRMED → PICKING → PACKED → DELIVERED` , cancel
- **Convert to invoice** → asli sale banti hai (`Sales.create`) — stock, customer ledger, loyalty, accounting sab wahi flow, advance payment ke sath
- **Take Order** full-screen mobile pad:
 - bara search (name / code / barcode), bari `+` / `-` qty buttons
 - customer picker (walk-in bhi), advance, delivery date, note
 - neechay sticky totals + **Order save karein**

Order formulas: `lineBase = price × qty - line discount` , bill discount lines par proportionally allocate (bilkul `Sales.create` jaisa), `total = subtotal - discount + tax` , `balance = total - advance` .

6. **POS payment polish**

- **Bill summary** payment modal ke upar (items, subtotal, discount, tax, total)

- **Fill** button — baqi amount ek click mein us method par
- **Quick chips** — `Exact`, round-up (`↑` with change), aur settings wale note buttons
- Wallet methods par asli gateway panel (section 2)

7. Verification (`bash tools/verify.sh`)

Gate	Kya check karta hai
<code>check</code>	Har <code>.gs</code> file ki syntax + HTML <code><script></code> blocks
<code>logic</code>	Real backend logic (stock, tax, ledger, accounting, wallet, orders) — 470 assertions
<code>ui</code>	Labels, accessibility, empty/undefined content
<code>smoke</code>	Har screen / tab / action — 245 checks
<code>tabs</code>	Har screen, sub-tab aur Settings sub-tab blank to nahi — 476 checks
<code>align</code>	Modal center / offcanvas right-edge alignment

Sab green hona chahiye: **ALL GATES GREEN** .